

Inventory Management using Near Field Communication (NFC)

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Abstract: *This research creates an NFC-inclusive website-based inventory management system that enhances tracking and management of stock. Scanning these tags allows users to obtain a specific product description without wasting time searching through the inventory. The solution is designed to enhance accuracy, efficiency, and user-friendliness for the convenience of the business through enhanced real time tracking and management of inventories.*

Keywords: *Near Field Communication, Inventory management, Product tracking, Automated inventory system, Internet of Things (IoT).*

I. INTRODUCTION

In today's commercial world, efficient inventory management is necessary for businesses who want to maintain an upper hand in the competition. Regular inventory systems rely on manual processes, which lead to inaccuracies. This increases labour costs, which lead to financial losses. As technologies continue to advance, businesses need innovative solutions that use modern communication technologies to enhance operational efficiency. One such technology is Near Field Communication (NFC). This technology enables seamless transfer of information between devices over short distances.

This research focuses on the development of an integrated system that combines NFC with a web-based interface. It also includes a MySQL database for efficient product information retrieval and real-time inventory tracking. This system summates the difficulties faced by a business to manage their stock by enabling the users to communicate with the products that are tap or coded through mobile devices using NFC (Near Field Communication). Once an NFC tag is read, the relevant information into a mobile application, such as product name, description, pricing brings useful information at hand and eliminates the usage of paper works.

In addition, the system is provided along with a web interface module that provides an option to the users called "Track Inventory" and users can view the levels of different items in the database. It also arms the companies with data to keep track of real-time availability and numbers of products, so they can make informed decisions. The use of

a MySQL database allows all products, inventory data to be stored securely, and it promotes data accuracy and representation. Integrating all of these components into one system helps streamline the inventory management process. This also helps improve the experience for users because it requires less manual data entry, which reduces the chances of human error.

II. LITERATURE REVIEW

Based on the power of data accuracy and automation, this study targets small to medium retail environments and the impact of their RFID technology on inventory control and viewing performance advantage information quickly. These results suggest a meaningful role of RFID on inventory and where the R^2 value, under 63.2% of multiple regression analysis, supports that RFID clearly improves inventory management (i.e. much better than human beings) [1]. In a large restaurant, the stock of perishable items is constantly taking place, for example, the items are getting out of date or in large quantities to waste, which brings problems such as stock, error, and waste are effectively made up of in the end with the development of a RFID-based inventory management system to Aiming to achieve storage-name optimization. This new idea aims to facilitate the process through on sight tracking and computerized data with an advantage of autotrace leading to a faster and efficient way to find your mistakes which will give you better insights on hashtag effectiveness and additional push for additional functionality and easier user interface in the future [2]. The RFID-based inventory management system using the ESP8266 microcontroller improves real-time tracking capability over local networks which greatly benefits small-scale businesses and warehouses. This integration is a tremendous time saver, reducing manual inventory counting time from three hours to less than half an hour, reducing human errors, provides better security and ensures no stock-outs and no overstocking [3]. This paper presents the design of an Inventory Management System (IMS) that aims to optimise operations for hardware stores in Indonesia by minimising inventory mismanagement and customer dissatisfaction. The Integrated Management System (IMS) enhances sales monitoring by automating processes,

notifying shop owners for re-orders, and providing an invoice generator along with a user-friendly interface. To assess current inventory practices and help businesses operating in the construction sector gain improved operational efficiency and better decision-making, the methodology is to make use of surveying construction professionals [4].

The Literature Review describes the transition from the tradition paper based manual inventory control system to automated systems, and the drawbacks of the Barcode system in terms of the risk of barcode damage and security. Highlights the benefits of NFC technology, including secure communication, ease of use and reliability, and refers to research that indicates its ability to improve both accuracy and operational efficiency. This lay the foundation for the NFC-based inventory management system for better management of electronic shops [5]. An inventory management system utilizes NFC technology to improve stocktaking efficiency at a steel company. The performance of this new system was compared with traditional manual methods. This study involved 11 participants who demonstrated a significant reduction in inventory check times with the NFC-enabled system ($M1 = 568.9$ seconds vs. $M2 = 759.4$ seconds). Statistical analysis confirmed that the new approach is faster and more reliable. This indicated that NFC technology can enhance the operational efficiency in inventory management [6]. This paper explores the evolution of smartphones and the rise of Near Field Communication (NFC). as an important technology for mobile payments and inventory control. It introduces an NFC-based inventory control system which enhances facility management and overcomes the problems of traditional methods. By using Android smartphones, the system enables real-time data transmission. This offers reduced manual stock-taking time which enhances the operational efficiency [7].

This paper explores the shift from standalone electronics to multifunctional, networked devices. This highlights the NFCIP-1 protocol for secure and effortless communication between devices. By enabling quick data exchange through proximity, NFC simplifies network creation without the difficulties of methods like Bluetooth or Wi-Fi [8]. The paper discusses three operational modes of NFC—reader/writer, peer-to-peer, and card emulation. It also discusses their roles in data exchange. The authors present a system where NFC tags act as wireless switches to control appliances via smartphones. These are supported by a smart extension box. Utilizing NFC Tools Pro and IFTTT applications for task automation, the results show effective device identification from a distance. This shows that NFC's can be used for smart home solutions [9].

III. METHODOLOGY

In addition, this study assists in enhancing the architecture of an all-encompassing system that incorporates NFC and web based interface. It is done so by embedding a MySQL within the system to facilitate product details and inventory management. Generally, the system can be divided into three main parts: the NFC based mobile application, the web-based system, and the MySQL database. The main objective of the project architecture is to ensure that interactions between the components are effective. This provides the efficient capabilities of fast reading and writing of the NFC labels as well as being able to manage inventory in a smooth way.

Every NFC tag consists a code which is unique and a product that has other associated information. These include product name, product description, product details and product pricing. Such kind of information is first encoded on the tags to provide additional security measures. The process of encoding this information makes use of symmetric encryption algorithms like AES. When using AES, a unique encryption key is created for each product. This digit is well protected in the database and is not fixed on the NFC tag. Rather, encrypted data is all that is fixed. This prevents the intrusion into the internal parts of the NFC tags where the actual content is stored. The encryption takes place via NFC writer app or hardware which does support this. Once the NFC tags are created and signed, it can be used along with the mobile app. It is developed with HTML, CSS, and JavaScript. This allows the users to interface with the system. It plugs into a backend language like PHP to take care of server-side logic. When an NFC tag is scanned, the mobile app decrypts the data from the tag and relays it to the web server. The server decrypts the data with the cached encryption key, and then the product details can be displayed to the user. In the web application, there is "Track Inventory" button, from where you can see what the product is available and how many it has. It accomplishes this through easy interface with MySQL database where product data and inventory are stored.

A. System level block diagram -

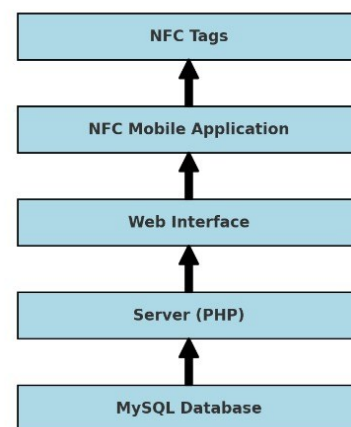


Fig. 1. System Level Block Diagram

In Fig. 1, MySQL database plays a role, in the system by storing product details and handling inventory information efficiently through tables that store various product attributes like ID number, name, description, price, quantity and encryption key. It also keeps track of real time availability data for products. The web application constantly updates the database whenever changes are made to inventory due, to sales or restocking operations. It makes sure that the details shown on both the application and web platform stay correct.

Checking how different NFC tags can be read and written is important to make sure the mobile app can correctly identify and handle them properly. Encrypting and decrypting information helps confirm that the right product details are accessed without revealing any data. To ensure data precision we compare the decrypted product details shown in the app and web interface with the information stored in the MySQL database.

IV. RESULTS AND DISCUSSION

The NFC Tagging system has advanced inventory management systems beyond tracking and managing products in the conventional way. End users can simply link a web page of the product to the NFC tag and check important details like cost, stock status, and features of the product. It minimizes the need for taking stock physically. This is both time-saving and effort-saving. NFC Readers are much cheaper than other technologies such as RFID.

These are built in most contemporary phones which makes the system less expensive. Even though NFC works in shorter distances than RFID, it is more suitable for places where prompt and localized inventory check is required. This system enhances security with the help of a special feature which performs an automatic overall checking system and determines likelihood of alteration of data associated with the product, which means the businesses do not have to worry in the first place. The system noticeably enhances inventory management and enables better operational flows. By improving the tracking of products, companies can control their inventories and respond to customers more rapidly and with more accurate information.

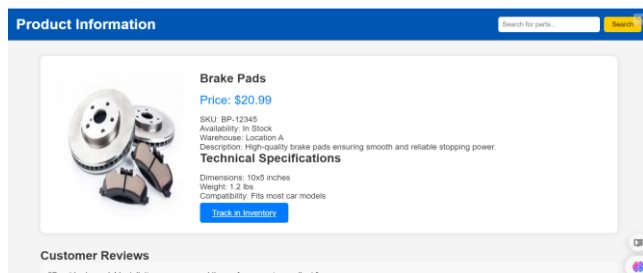


Fig 2. Product Information

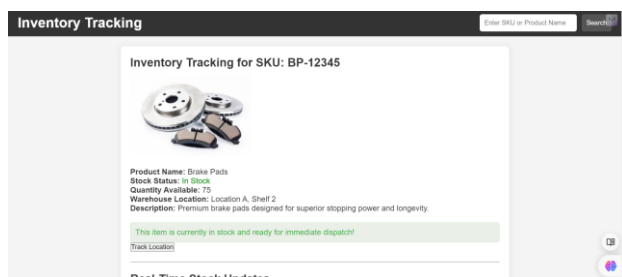


Fig 3. Inventory Tracking

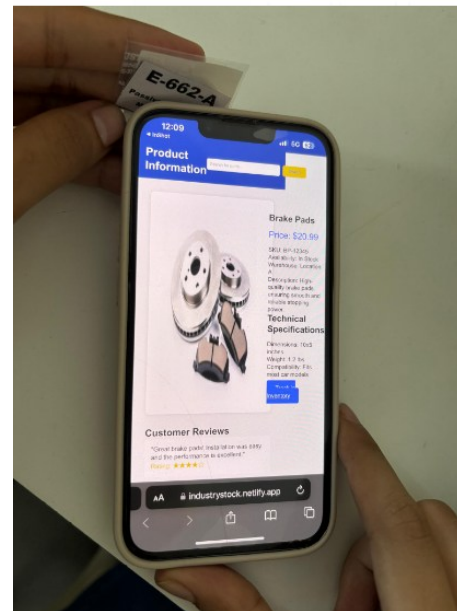


Fig 4. NFC Tag Detection

Fig. 2 shows the information of the product when the NFC tag is detected. It gives the overview of the product. Fig. 3 shows the stock of the product in the inventory. This helps in tracking the inventory. Fig 4. shows how the NFC tag is scanned by the mobile.

IV. CONCLUSION

The development of web-based systems owning a MySQL database and integrated with NFC technology may represent a high level for the universality of the inventory management systems. With the adoption of NFC for fast scanning of the tags, product detail retrieval and stock take becomes easy, this in turn reduces human mistakes and increasing the effectiveness of work. In the faster and safer domains, the system is useful and dependable since private information being encrypted is kept private. The mobile application, database and web interface are able to communicate effectively, resulting in efficient management of goods in stock, and is convenient and expandable for many scopes of usage. To sum up, the new system developed will improve the accuracy, effectiveness and dependability of an inventory system while solving the challenges that come with usage of manual systems as well as data protection systems, is an application that is likely to have great impact in industries like retail, logistics, healthcare, manufacturing among others where controlling the costs of economies looms larger for the attainment of corporate goals.

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